Petri Net

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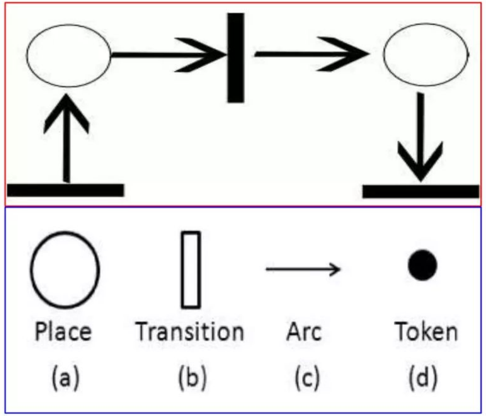
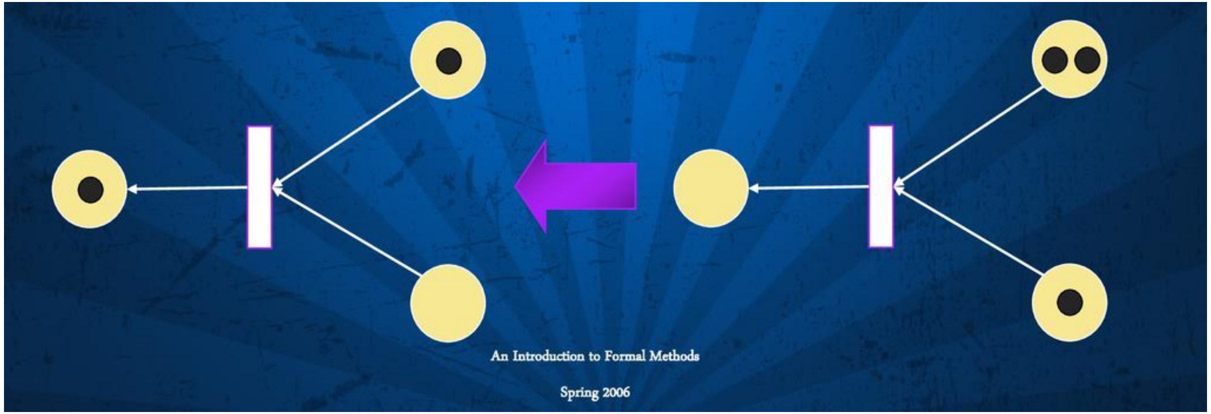
A Petri net is a 5-tuple, $PN = (P, T, F, W, M_0)$ where:

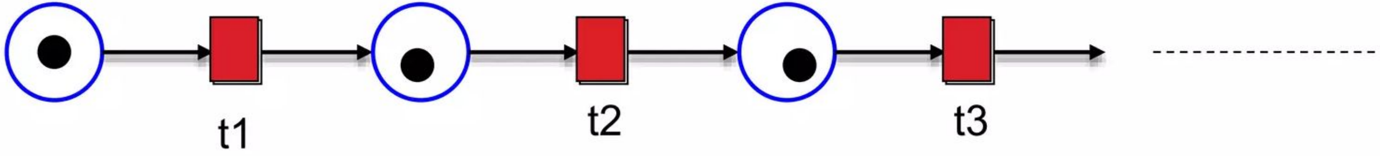
$P = \{p_1, p_2, \cdots, p_m\}$ is a finite set of places,
$T = \{t_1, t_2, \cdots, t_n\}$ is a finite set of transitions,
$F \subseteq (P \times T) \cup (T \times P)$ is a set of arcs (flow relation),
$W: F \rightarrow \{1, 2, 3, \cdots\}$ is a weight function,
$M_0: P \rightarrow \{0, 1, 2, 3, \cdots\}$ is the initial marking,
$P \cap T = \varnothing$ and $P \cup T \neq \varnothing$.

A Petri net structure $N = (P, T, F, W)$ without any specific initial marking is denoted by N.

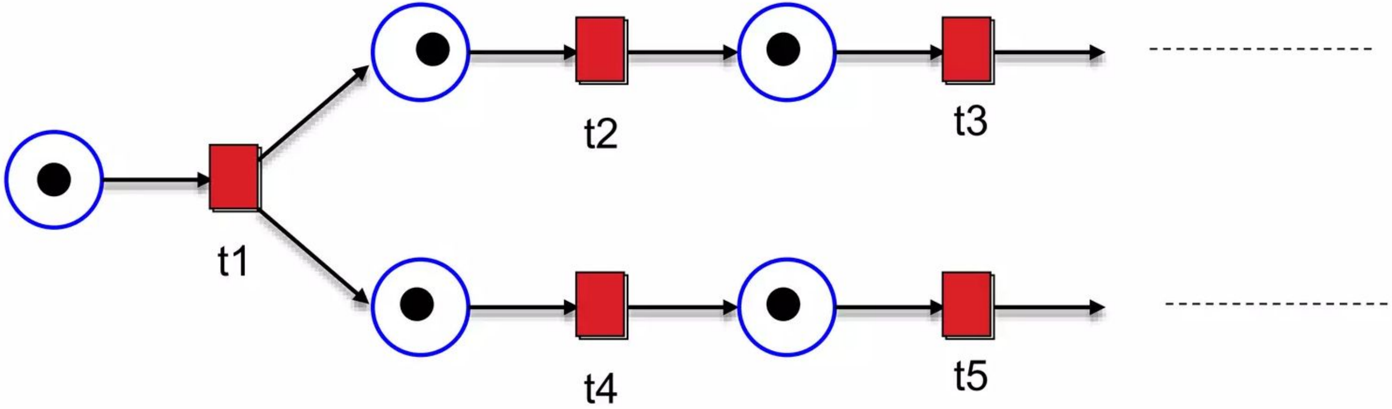
A Petri net with the given initial marking is denoted by (N, M_0).



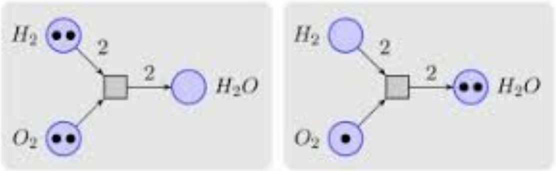


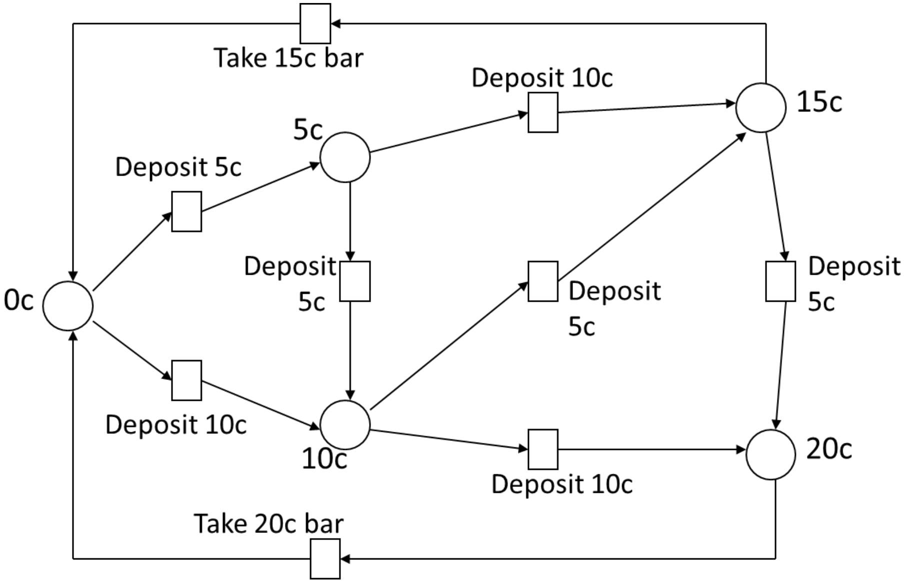


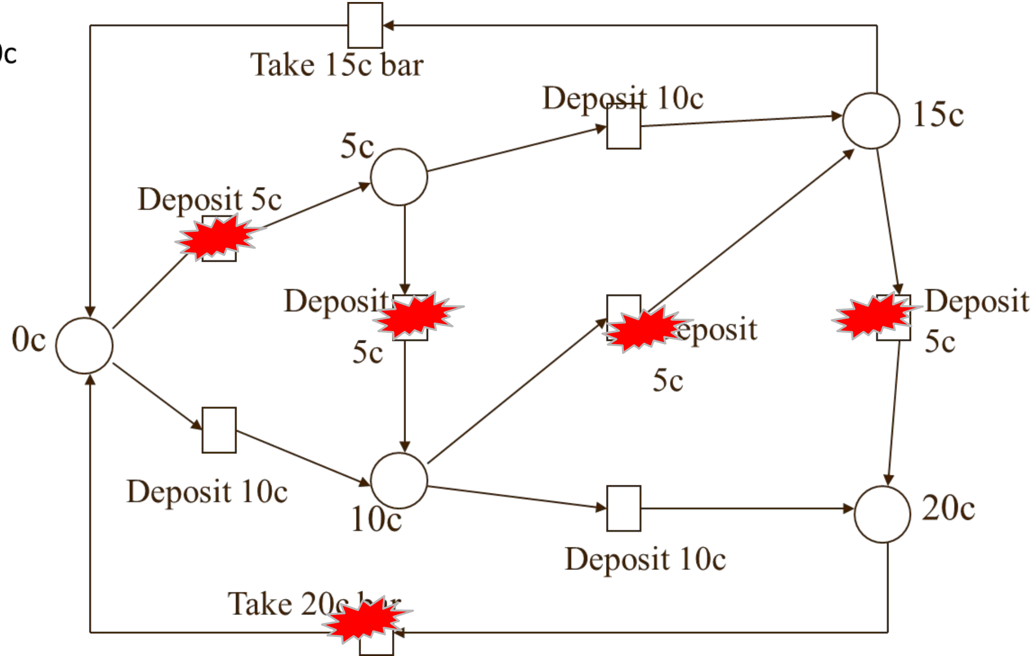
(A sequence Structure)



(Concurrent Structure)




Take 15c bar
Deposit 10c
15c
5c
Deposit 5c
Deposit
5c
Deposit
5c
Deposit
5c
0c
Deposit 10c
10c
Deposit 10c
20c
Take 20c bar


Take 15c bar
Deposit 10c
15c
5c
Deposit 5c
Deposit
5c
Deposit
5c
0c
5c
Deposit 10c
10c
Deposit 10c
20c
Take 20c

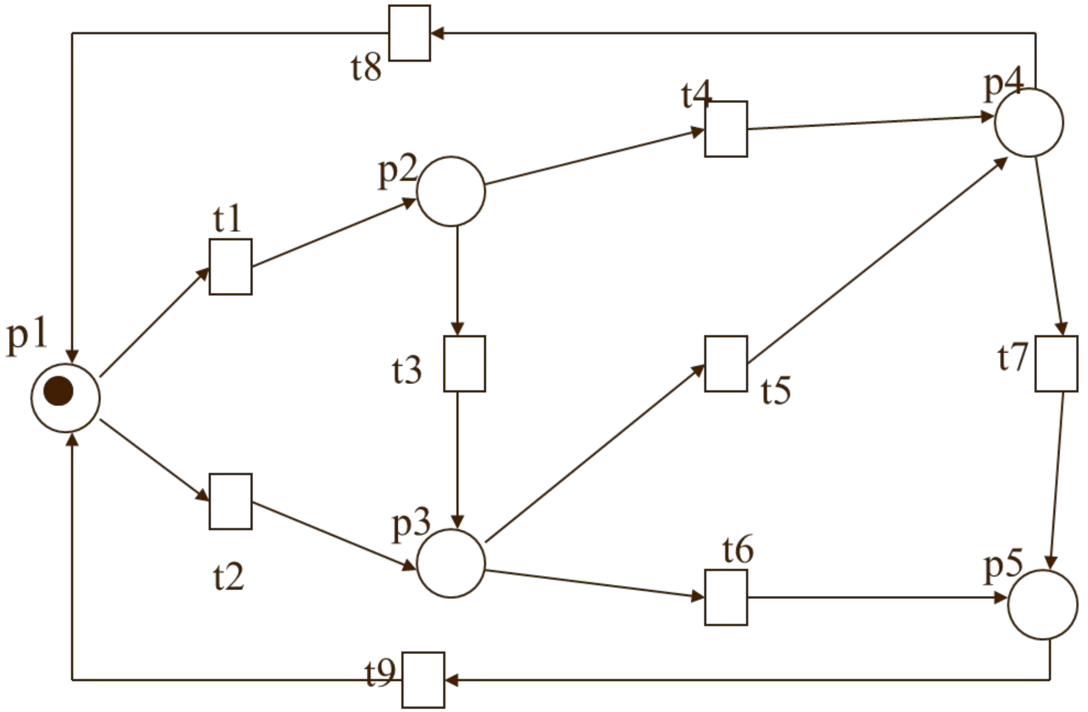
$M0 = (1,0,0,0,0)$

$M1 = (0,1,0,0,0)$

$M2 = (0,0,1,0,0)$

$M3 = (0,0,0,1,0)$

$M4 = (0,0,0,0,1)$



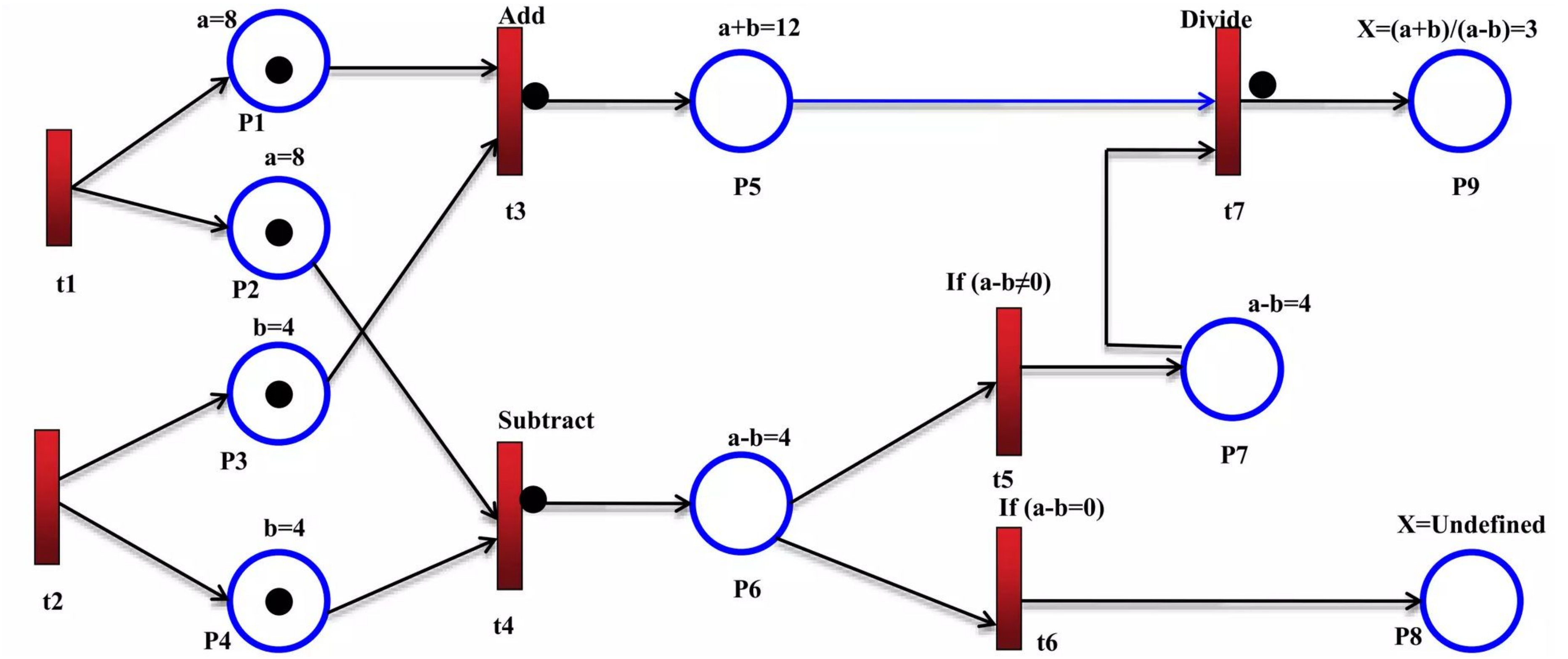
Initial marking:M0

$$M0 \xrightarrow{t1} M1 \xrightarrow{t3} M2 \xrightarrow{t5} M3 \xrightarrow{t8} M0 \xrightarrow{t2} M2 \xrightarrow{t6} M4$$

Stochastic Petri Net (SPN)

transitions happen at random times

$X=(a+b)/(a-b)$



a=8
P1
a=8
P2
b=4
P3
b=4
P4
t1
t2
Add
t3
Subtract
t4
a+b=12
P5
a-b=4
P6
If (a-b≠0)
t5
If (a-b=0)
t6
a-b=4
P7
Divide
t7
X=(a+b)/(a-b)=3
P9
X=Undefined
P8

Petri Net Components for Emotional Modeling

Places (Emotional States)

Represent distinct emotional conditions like Happiness, Sadness, Anger, or Calm.

Transitions (Triggers)

Events or stimuli that provoke changes in emotional state (e.g., news or loss).

Tokens (Intensity)

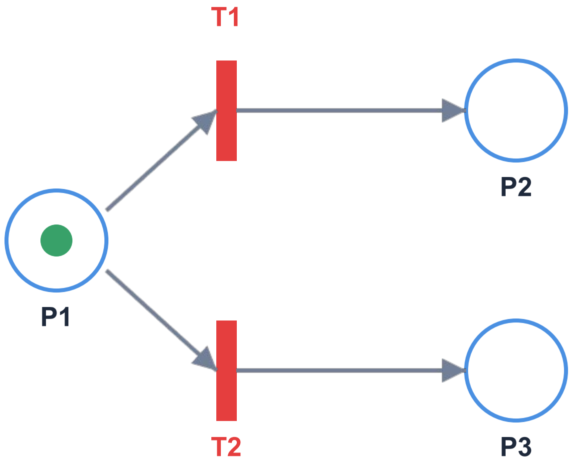
Represent the strength of an emotion. Movement indicates a shift in emotional state.

Arcs (Relationships)

Connect places and transitions, defining the causal link between emotions and events.

Emotional State Modeling: Feedback Loop

This Petri Net model illustrates how an individual's emotional state transitions between Neutral, Happiness, and Sadness based on Positive and Negative Feedback events.



Components

Places (Emotional States)

- P1: Neutral (Starting state)
- P2: Happiness
- P3: Sadness

Transitions (Feedback Events)

- T1: Positive Feedback (Praise, Success)
- T2: Negative Feedback (Criticism, Failure)

Tokens (Current State)

Represent the active emotional state within the model.

Emotional State Modeling: System Steps

1. Initial State

The system starts with a token in the "Neutral" place P1.

2. Positive Feedback

Positive Feedback T1 moves a token from "Neutral" P1 to "Happiness" P2.

3. Negative Feedback

Negative Feedback T2 moves a token from "Neutral" P1 to "Sadness" P3.

4. Emotional Convergence

Positive: Token accumulates in "Happiness" with more feedback.

Negative: Token shifts to "Sadness" upon negative feedback.

5. Concurrency & Conflict

The model extends to simultaneous feedback, potentially leading to token splits or complex states like **"Ambivalence."**

Petri Net: Structural Logic

Defining the foundational components and interaction arcs for emotional state modeling.

Places (States)

P1: Neutral Baseline emotional state

P2: Happiness Positive emotional target

P3: Sadness Negative emotional target

Transitions (Feedback)

T1: Positive Feedback Trigger for happiness shift

T2: Negative Feedback Trigger for sadness shift

Arcs (Flow Logic)

`P1 -> T1 -> P2`
Neutral to Happiness via Positive

`P1 -> T2 -> P3`
Neutral to Sadness via Negative

Operational Scenarios & Extensions

Logic Scenario

- **Initial:** Start with a token in P1 (Neutral).
- **Positive Feedback (T1):** Token moves from P1 to P2 (Happiness).
- **Negative Feedback (T2):** Token shifts from P1 or P2 to P3 (Sadness).

Intensity Modeling

Tokens carry weights representing the **magnitude** of emotional responses.

Complex Emotions

Integration of states like **"Anxiety"** or **"Relief"** via multi-event transitions.

Temporal Dynamics

Time-based nets capture **delay** or **decay** of emotional states over time.

Process Modelling and Creating Predictive Models of Sensory Networks Using Fuzzy Petri Nets

Physiological Data for Emotion Recognition

Physiological responses (like heart rate, skin conductivity) and **facial expressions** are rich sources of information for recognizing emotions.

Data Acquisition & IoT

Non-invasive IoT devices: Smart bands and smartphones provide a seamless way to track metrics.

Microcontroller-based modules: Arduino and Raspberry Pi allow for customized, real-time sensing.

Dataset & Sensory Network

A **complex sensory network** is formed to capture a comprehensive dataset of physiological states.

- Real-time data collection
- High-fidelity state tracking
- Multi-modal information fusion

Modeling Emotional States with Petri Nets

To handle the complexity and uncertainty of emotional states, **Fuzzy Petri Nets (FPNs)** are introduced. FPNs integrate **fuzzy logic** into Petri nets.

Dealing with Ambiguity

FPNs are useful for dealing with **ambiguous, uncertain data**—like emotions, where it is often impossible to pinpoint an exact state. They provide a robust framework for mapping physiological signals to fluid emotional transitions.

Transition Logic in Emotion Modeling

In this model, **places** represent physiological states, while **transitions** signify changes in emotional states triggered by data thresholds.

Dynamic State Shifting

When a threshold of physiological tokens is reached, the transition fires, moving the system to a new state.

Example:
Moving from a **neutral** state to **stressed** based on heart rate and skin conductivity readings.

Continuous Prediction

The system processes real-time data to anticipate future emotional trajectories.

- Real-time data processing
- Historical data integration
- Predictive state modeling

Inputs: Physiological States

Galvanic Skin Response (GSR)

- Measures skin conductivity via sweat glands in palms and soles.
- Linked to arousal, reflecting emotional and cognitive activity.

Electrocardiography (EKG)

- Measures heart's electrical activity.
- Computes **HR, IBI,** and **HRV**.
- Differentiates emotions; highly sensitive to emotional states.

Heart Rate (HR)

Provides direct insight into emotional and physiological states.

Primary Use:
Identifying levels of stress or excitement.

Electromyography (EMG)

- Detects muscle activity via surface voltages during contraction.
- Distinguishes between positive/negative emotions based on facial muscle activity.

Outputs: Emotional States

The output of the model is the **classified emotional state** of the user, inferred from input physiological data. Predicted states include:

😃 Positive Emotions

Detected through physiological markers indicating a positive affective state:

- Higher EMG activity in cheek muscle regions
- Higher EMG activity in periocular muscles
- Increase in heart rate (HR)
- Increased skin conductance

☹️ Negative Emotions

Identified by markers associated with stress or negative affect:

- Higher EMG activity in the brow region
- Lower heart rate readings
- Specific variations in GSR levels

Predicting Emotional States

The focus is on the outputs of the model — the **emotional states** — and the psychological frameworks that aid in classifying these emotions.

The aim is to predict basic emotional states based on physiological inputs using **Petri nets**.

Model Objective

Russell's Circumplex Model

A widely accepted model for classifying emotional states by plotting them on two primary dimensions.

Emotions are positioned in a circular structure around a neutral center.

Arousal

Vertical axis measuring the intensity of experience.

Valence

Horizontal axis determining if an emotion is pleasant.

Model Examples:

- High arousal + Positive valence = **Excitement**
- Low arousal + Negative valence = **Sadness**

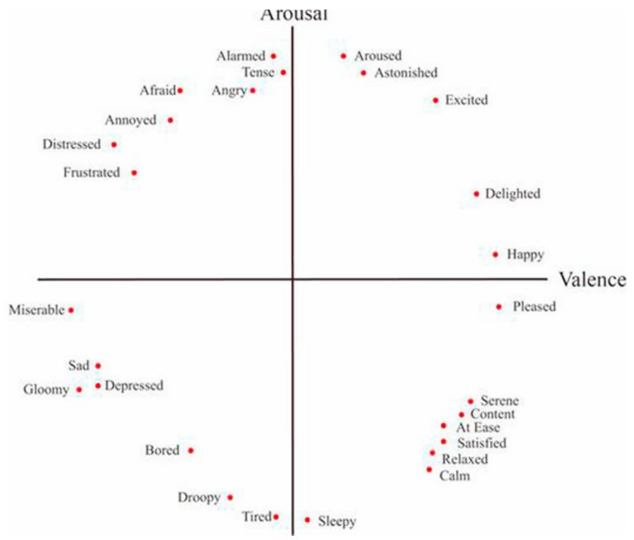


Fig. 1. Russel's circumplex model

Fig 1. Russell's Circumplex Model Illustration

Lövheim's Cube Model

A three-dimensional structure linking emotions to the levels of three primary neurotransmitters:

Dopamine

Noradrenaline

Serotonin

Each corner represents an emotional state (e.g., **fear**, **anger**, **happiness**) based on these neurotransmitter levels.

Intensity varies as you move from the **corners** (extreme) towards the **neutral center**.

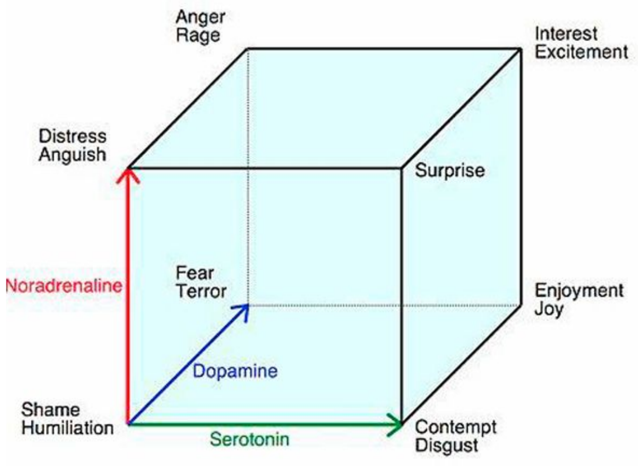


Fig. 2. Lövheim's cube

Fig 2. Lövheim's Cube Model Illustration

Fuzzy Logic in Emotion Prediction

Because emotions exist on a spectrum and may be hard to classify exactly, **fuzzy logic** handles uncertainties in emotional state prediction. **Fuzzy rules** (IF-THEN) trigger transitions within the Petri net for classification.

Rule Example 01

IF **GSR** and **HR** are high...

THEN Result: STRESS

Rule Example 02

IF **EKG** (Moderate HR) & **EMG** active...

THEN Result: HAPPINESS

Fuzzy logic allows the system to map objective physiological data into subjective emotional spectrums with high precision.

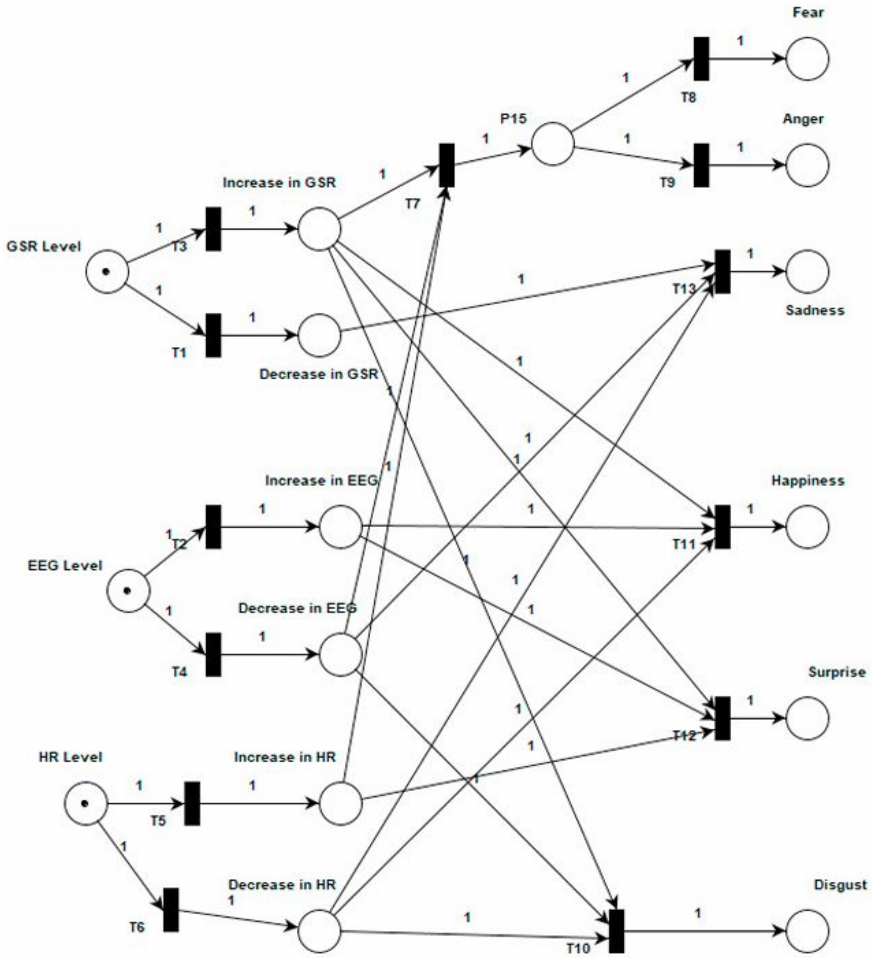


Fig. 3. The created conceptual model in Petri nets

Future Directions

The emotion recognition framework is poised for expansion, integrating more complex physiological markers and nuanced emotional classification.

Facial Recognition

Integrating visual cues for multi-modal accuracy.

Body Temperature

Thermal imaging for stress and arousal detection.

Cardiovascular Measures

Advanced PPG and ECG heart rate variability analysis.

Moreover, classification can be expanded to include not only basic emotions but potentially a broader range of complex emotional states.